

WEEK 4

Statistics, Data Visualization & Exploratory Data Analysis

Student	Tiwari Aman Dhananjay
Assignment	Week 4 — Statistics, Data Visualization & EDA
Duration	1 Week (7 Days)
Marks	100
Tools	Python • Pandas • NumPy • Matplotlib • Seaborn

Practical Data Analytics Assignment

1. Assignment Overview

This project applies descriptive statistics, probability, correlation, outlier detection, Matplotlib and Seaborn visualization, data inspection, cleaning, EDA and business recommendations to a practical sales dataset.

Task	Topic	Marks
1	Mean, Median & Mode	10
2	Variance & Standard Deviation	10
3	Correlation & Probability	10
4	Outlier Detection	10
5	Matplotlib Visualization	15
6	Seaborn Visualization	15
7	EDA Inspection & Cleaning	15
8	EDA Correlation & Insights	10
9	Business Recommendations	5

2. Statistical Results

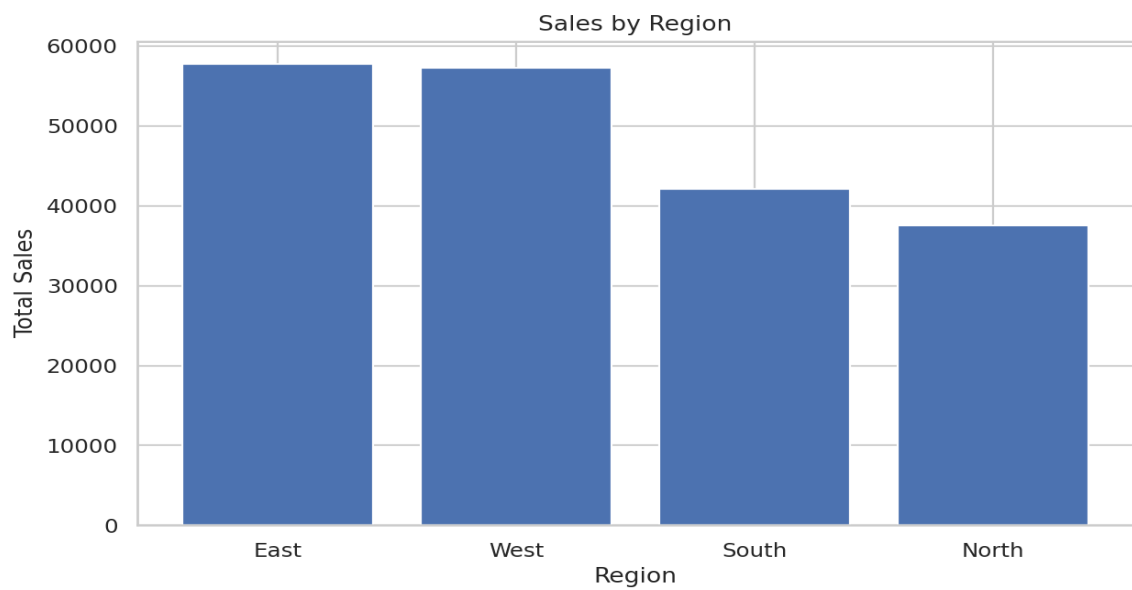
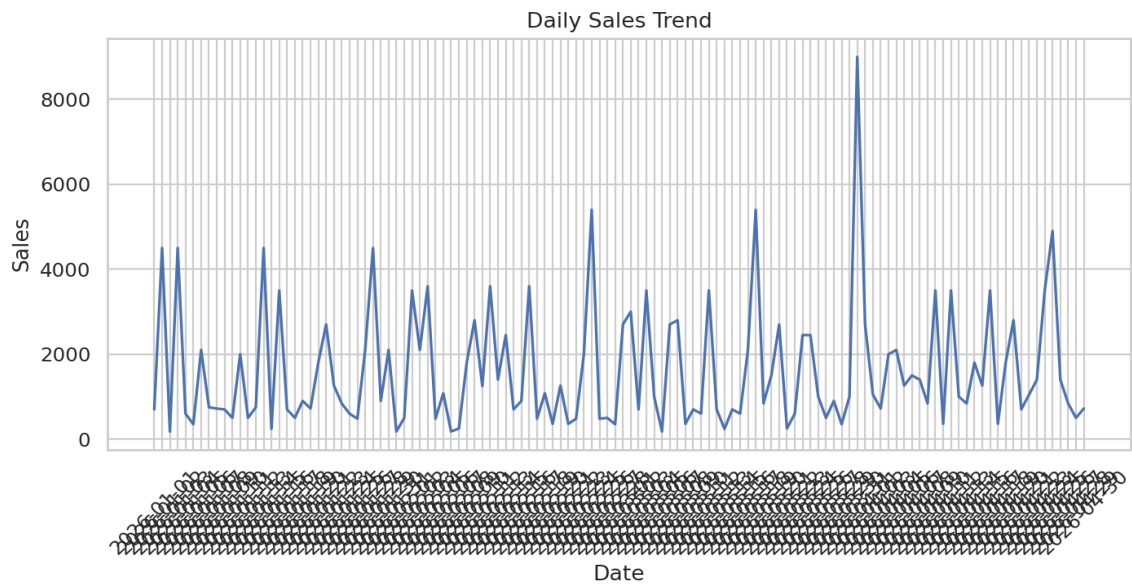
Metric	Result
Mean Sales	1,621.92
Median Sales	1,000.00
Mode Sales	700.00
Variance	2,127,634.11
Standard Deviation	1,458.64
Quantity-Sales Correlation	0.50
P(Sales > Median)	49.17%
IQR Outliers	3

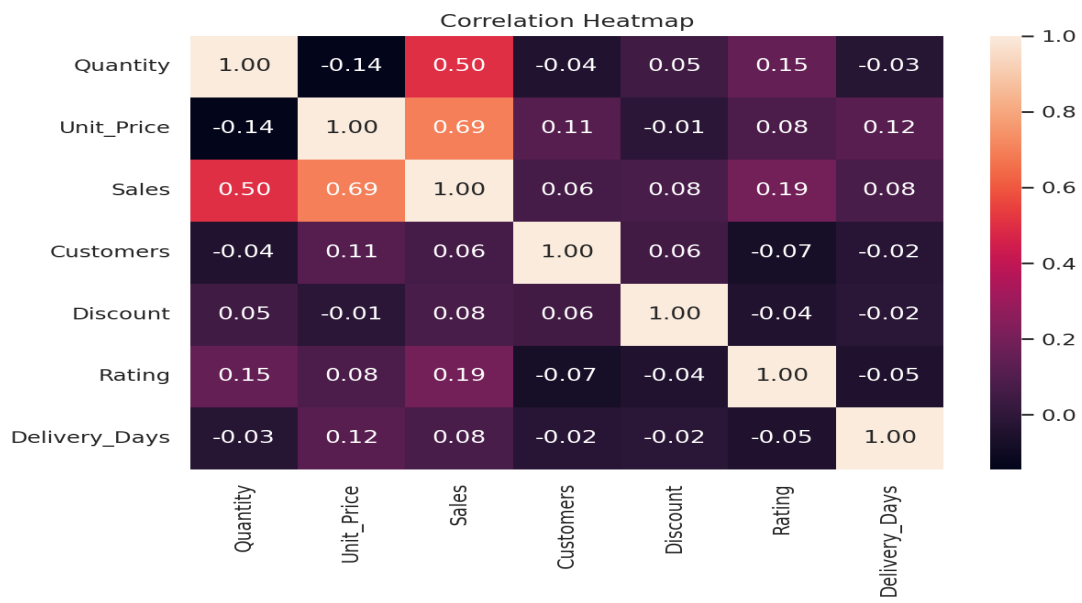
Interpretation

Mean is the average, median is the middle value after sorting, and mode is the most frequent value. Variance and standard deviation describe spread. The positive quantity-sales correlation indicates that larger quantities tend to be associated with higher sales.

3. Visualization Work

The project includes all five required Matplotlib charts and all four required Seaborn visualizations. The PNG files are stored in the outputs folders.





4. EDA — Inspection, Cleaning & Insights

Before cleaning, the dataset contained 122 rows and 10 columns, with 5 missing cells and 2 duplicate records. Categories were standardized, missing numerical values were filled with medians, duplicates were removed, and the extreme quantity was capped before recalculating sales.

Metric	Value
Rows before	122
Columns	10
Missing cells	5
Duplicates	2
Rows after	120
IQR outliers	3

Key Insights

- East generated the highest total sales at 57,740.
- Monitor contributed the highest product-level sales at 59,040.
- Quantity and Sales have a 0.50 correlation, indicating a positive relationship.
- Average customer rating is 3.97/5 and average delivery time is 4.58 days.
- IQR analysis identified 3 potential sales outlier(s).

5. Business Recommendations

- Prioritize inventory and targeted campaigns in the East region because it has the highest sales contribution.
- Review demand and promotion strategy for Monitor, the highest-sales product.
- Investigate high-value outlier transactions before using the data for forecasting or predictive modeling.

6. Submission Checklist

- Python source code included
- Jupyter notebook included
- Raw and cleaned datasets included
- Matplotlib outputs included
- Seaborn outputs included
- EDA tables and cleaned data included
- README and requirements included
- Genuine screenshots should be added to screenshots/

7. Conclusion

The project demonstrates a complete introductory analytics workflow: inspect data, clean inconsistencies, calculate descriptive statistics, study relationships, identify outliers, visualize patterns and convert findings into data-driven business recommendations.

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